



# Outline

Anatomy of a NN

Design choices

- Activation function
- **Loss function**
- Output units
- Architecture



# Loss Function

Likelihood for a given point:

$$p(y_i|W; x_i)$$

Assume independency, likelihood for all measurements:

$$L(W; X, Y) = p(Y|W; X) = \prod_i p(y_i|W; x_i)$$

Maximize the likelihood, or equivalently maximize the log-likelihood:

$$\log L(W; X, Y) = \sum_i \log p(y_i|W; x_i)$$

Turn this into a loss function:

$$\mathcal{L}(W; X, Y) = -\log L(W; X, Y)$$



# Loss Function

Do not need to design separate loss functions if we follow this simple procedure

## Examples:

- Distribution is **Normal** then likelihood is:

$$p(y_i|W; x_i) = \frac{1}{\sqrt{2\pi^2\sigma}} e^{-\frac{(y_i - \hat{y}_i)^2}{2\sigma^2}}$$

**MSE**

$$\mathcal{L}(W; X, Y) = \sum_i (y_i - \hat{y}_i)^2$$

- Distribution is **Bernoulli** then likelihood is:

$$p(y_i|W; x_i) = p_i^{y_i} (1 - p_i)^{1-y_i}$$

**Cross-Entropy**

$$\mathcal{L}(W; X, Y) = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$$